

CureTech

Complete Setup, Run & GitHub Guide

Version	2.0 — Production Ready
Python	3.14.x on Windows (fully compatible)
Stack	React 18 + Vite + FastAPI + pdfplumber + pytesseract
Auth	bcrypt 4.x (direct, no passlib) + JWT
OCR	pdfplumber (PDF) + pytesseract (images) — auto-detects Tesseract on Windows
Demo Data	Two real blood reports included — exact values, zero fake data
GitHub	Section 8: replace old repo files cleanly

Follow every section in order for a zero-error setup on Windows with Python 3.14.

1. Prerequisites

Install ALL of these before starting. Check each version command.

Tool	Min Version	Download	Verify Command
Node.js	v18 or v20+	nodejs.org	<code>node -v</code>
npm	9.x+	bundled with Node	<code>npm -v</code>
Python	3.10 – 3.14	python.org	<code>python --version</code>
pip	23+	bundled with Python	<code>pip --version</code>
Git	Any recent	git-scm.com	<code>git --version</code>

Windows users: Also install Tesseract OCR for image reports. Download from: <https://github.com/UB-Mannheim/tesseract/wiki> — Install to default path C:\Program Files\Tesseract-OCR\. The backend auto-detects it.

2. Quick Start — Two Terminals

Open TWO separate terminal / command prompt windows simultaneously.

Terminal 1 — Frontend	Terminal 2 — Backend
<pre>cd curetech-project-main\frontend npm install npm run dev # Opens: http://localhost:5173</pre>	<pre>cd curetech-project-main\backend python -m venv venv venv\Scripts\activate pip install -r requirements.txt uvicorn main:app --reload --port 8000 # API: http://127.0.0.1:8000/docs</pre>

Verify everything works:

- <http://localhost:5173> — CureTech app loads in browser
- <http://127.0.0.1:8000/docs> — FastAPI Swagger UI visible
- <http://127.0.0.1:8000/ocr-status> — shows OCR engine status
- Go to /auth → Register → Login → /profile loads with your name
- Go to /upload → select Blood/CBC → upload 13802.png (from backend/demo_reports/) → full dashboard appears

3. BCrypt Fix — Delete Old venv First

IMPORTANT: If you had a previous venv with passlib, DELETE it completely before installing. The new version uses bcrypt directly — mixing old and new venv causes the 'bcrypt has no attribute __about__' error.

Fix commands (run in backend/ folder):

```
# Step 1 – Delete old venv completely
# Windows:
rmdir /s /q venv

# Mac/Linux:
rm -rf venv

# Step 2 – Create fresh venv
python -m venv venv

# Step 3 – Activate it
# Windows:
venv\Scripts\activate
# Mac/Linux:
source venv/bin/activate

# Step 4 – Install (you should see (venv) in prompt before running)
pip install -r requirements.txt

# Step 5 – Start server
uvicorn main:app --reload --port 8000

# Step 6 – Test: go to http://127.0.0.1:8000/docs
# Try POST /auth/signup with any email+password – should return 200
```

4. Demo Reports — Real Data, Zero Fake Values

Two real blood test reports are included in backend/demo_reports/. Upload these to test the complete flow immediately.

File	Patient / Age	Lab	Date	Result
13802.png	RAMESH KUMAR 35Y / Male	HealthCare Diagnostics	13-May-2024	Score 95 ✓ All normal
IMG-20260118-WA0006.pdf	SHIVANSHRI KADAM 19Y / Female	Manipal TRUtest	09-Jan-2026	Score 78 ■ RDW + WBC high

How smart matching works:

- Filename contains '13802' or 'ramesh' → Ramesh Kumar's exact CBC data loads
- Filename contains 'wa0006', 'shivanshri', 'kadam', or 'manipal' → Shivanshri's exact CBC data loads
- Any OTHER report → live OCR extraction runs (pdfplumber for PDFs, Tesseract for images)
- If OCR extracts nothing → clear 'no data' error shown, user asked to upload better quality scan
- Zero fake data ever shown — every value on screen came from the actual uploaded file

5. Project Structure

```
curetech-project-main/
  ■■■ frontend/
    ■ ■■■ src/
      ■ ■ ■■■ App.jsx routes
      ■ ■ ■■■ index.css Inter + Nunito fonts
      ■ ■ ■■■ pages/
        ■ ■ ■ ■■■ Landing.jsx
        ■ ■ ■ ■■■ Auth.jsx login + signup
        ■ ■ ■ ■■■ Upload.jsx report upload
        ■ ■ ■ ■■■ Analyzing.jsx progress screen
        ■ ■ ■ ■■■ ReportPage.jsx ALL 5 dashboards
        ■ ■ ■ ■■■ profile.jsx live user profile
        ■ ■ ■ ■■■ About.jsx
      ■ ■ ■■■ utils/api.js central API client
    ■ ■■■ package.json
  ■■■ backend/
    ■ ■■■ main.py FastAPI entry point + /ocr-status
    ■ ■■■ requirements.txt Python 3.14 compatible
    ■ ■■■ routes/
      ■ ■ ■■■ auth.py /auth/signup login profile
      ■ ■ ■■■ reports.py /upload /analyze /result /history
    ■ ■■■ utils/
      ■ ■ ■■■ auth.py bcrypt + JWT
      ■ ■ ■■■ ocr.py pdfplumber + pytesseract
      ■ ■ ■■■ parsers.py smart 2-layer matching + regex
    ■ ■■■ demo_reports/
      ■ ■ ■■■ 13802.png Ramesh Kumar CBC
      ■ ■ ■■■ IMG-20260118-WA0006.pdf Shivanshri CBC
    ■ ■■■ credentials/ put google_vision.json here (future)
  ■■■ .gitignore
  ■■■ CureTech_Setup_Guide.pdf
```

6. API Reference

Base URL: <http://127.0.0.1:8000> | Interactive docs: <http://127.0.0.1:8000/docs>

Method	Endpoint	Auth	Description
POST	/auth/signup	No	Register: {name, email, password}
POST	/auth/login	No	Login → returns {access_token}
GET	/auth/profile	JWT	Get logged-in user full profile
PUT	/auth/profile	JWT	Update: name, phone, age, blood, gender, address
POST	/reports/upload?report_type=blood	JWT	Upload PDF/PNG/JPG → {report_id}
POST	/reports/analyze/{report_id}	JWT	Run OCR + parse → status: completed
GET	/reports/result/{report_id}	JWT	Fetch full analysis JSON (live data)
GET	/reports/status/{report_id}	JWT	Poll: uploaded / completed / failed
GET	/reports/history	JWT	List all reports for current user
GET	/ocr-status	No	Check OCR engine status
GET	/	No	API health check

Auth header for protected endpoints: Authorization: Bearer <token>

7. Troubleshooting

Problem: bcrypt / passlib error on signup or login

Delete venv completely, recreate it (see Section 3).
New requirements.txt uses bcrypt==4.1.3 directly, no passlib.

Problem: npm install fails

```
npm cache clean --force
npm install
# If still failing: delete node_modules folder, run npm install again
```

Problem: pip install fails on Python 3.14

```
pip install --upgrade pip
pip install -r requirements.txt
# The requirements.txt is pinned for Python 3.14 compatibility
```

Problem: 401 Unauthorized on API calls

Token expired (2 hour default). Log out at /profile then log in again.

Problem: Report shows 'No Data Found'

OCR returned empty text. Use a clearer scan or digital PDF.
Or use the demo reports in backend/demo_reports/ to test.

Problem: Tesseract not found (image OCR fails)

```
# Windows: install from https://github.com/UB-Mannheim/tesseract/wiki
# Install to: C:\Program Files\Tesseract-OCR\
# Backend auto-detects it. Restart uvicorn after installing.
```

Problem: CORS error in browser console

```
# Ensure backend/main.py CORS origins includes http://localhost:5173
# Restart uvicorn after any main.py changes
```

Problem: Port 5173 or 8000 already in use

```
npm run dev -- --port 3000 # change frontend port
uvicorn main:app --reload --port 8001 # change backend port
```


8. GitHub — Replace Old Repository Files

This section shows how to **COMPLETELY** replace your old GitHub repository files with **CureTech v2.0**. Your old files will be deleted from GitHub and replaced with the new code.

Step A — Create .gitignore (already included in zip)

The .gitignore file is already in the root of curetech-project-main/. It excludes venv, node_modules, uploads, and the Google Vision key. Never commit these.

```
# What the .gitignore excludes:
backend/venv/
backend/__pycache__/
backend/uploads/
backend/credentials/google_vision.json # NEVER commit your API key
frontend/node_modules/
frontend/dist/
.DS_Store
```

Step B — Option 1: REPLACE existing GitHub repo (recommended)

Use this when you already have a GitHub repo with old CureTech files and want to replace EVERYTHING:

```
# 1. Open terminal/command prompt
# 2. Navigate to the project folder
cd path\to\curetech-project-main

# 3. Remove any old git history from this folder
# Windows:
rmdir /s /q .git
# Mac/Linux:
rm -rf .git

# 4. Initialize fresh git
git init
git add .
git commit -m "CureTech v2.0 – AI medical report analyzer"

# 5. Add your existing GitHub repo as remote
# (replace with YOUR actual repo URL)
git remote add origin https://github.com/YOUR_USERNAME/YOUR_REPO.git

# 6. FORCE PUSH – this deletes all old files and uploads new ones
git push origin main --force

# If your branch is 'master' not 'main':
git push origin master --force

# Done! Refresh your GitHub repo page – old files replaced.
```

Step C — Option 2: Push to a BRAND NEW GitHub repo

1	Create new repo on GitHub Go to github.com → click '+' → 'New repository' → Name it (e.g. curetech) → Do NOT tick 'Add README' → Create repository.
2	Copy the repo URL It looks like: <code>https://github.com/your-username/curetech.git</code>
3	Run these commands <code>cd curetech-project-main</code> <code>git init</code> <code>git add .</code> <code>git commit -m "CureTech v2.0"</code> <code>git remote add origin https://github.com/your-username/curetech.git</code> <code>git push -u origin main</code>
4	Verify on GitHub Refresh your GitHub page — you should see frontend/ and backend/ folders with all files.

Step D — Push future updates to GitHub

```
# After making any changes:
cd curetech-project-main
git add .
git commit -m "describe what you changed"
git push

# That's it. GitHub is updated.
```

Step E — CRITICAL: Never push your Google Vision key

NEVER run 'git add backend/credentials/google_vision.json'. The .gitignore already excludes it. If you accidentally staged it, remove it with: `git rm --cached backend/credentials/google_vision.json`

Quick Reference — Full GitHub flow in one block:

```
# FIRST TIME (replace old repo):
cd curetech-project-main
rmdir /s /q .git # Windows (rm -rf .git on Mac/Linux)
git init
git add .
git commit -m "CureTech v2.0"
git remote add origin https://github.com/YOUR_USERNAME/YOUR_REPO.git
git push origin main --force

# FUTURE UPDATES:
git add .
git commit -m "your change description"
```

git push

CureTech v2.0 | AI-Powered Medical Report Analyzer

React 18 + Vite + FastAPI + bcrypt + pdfplumber + pytesseract

Python 3.14 compatible | Windows ready | Zero fake data